

Authentication Guide

Authentication Flow

The login service validates user credentials using email and password. After successful verification, a JWT token is generated and attached to the response. Middleware checks token validity on every protected route. Session expiration is handled through token expiry timestamps.

Password Security

Passwords are hashed using bcrypt before being stored in the database. Authentication failures return unauthorized responses and prevent access to protected resources. Refresh tokens can be used to obtain new access tokens without forcing users to log in again.

Authorization

Role-based access control determines which resources a user can access. Admin users have elevated privileges while normal users can only access their own data. Unauthorized requests return HTTP 403 responses.

Machine Learning Basics

Machine learning models are trained using datasets. Neural networks learn patterns through backpropagation and gradient descent. Classification algorithms predict categories while regression models predict numerical values. Model performance is evaluated using accuracy, precision, recall, and F1 score.

Database Connections

Database connections are managed through connection pools. Queries are optimized using indexes to improve performance. Transactions ensure consistency and prevent data corruption.